

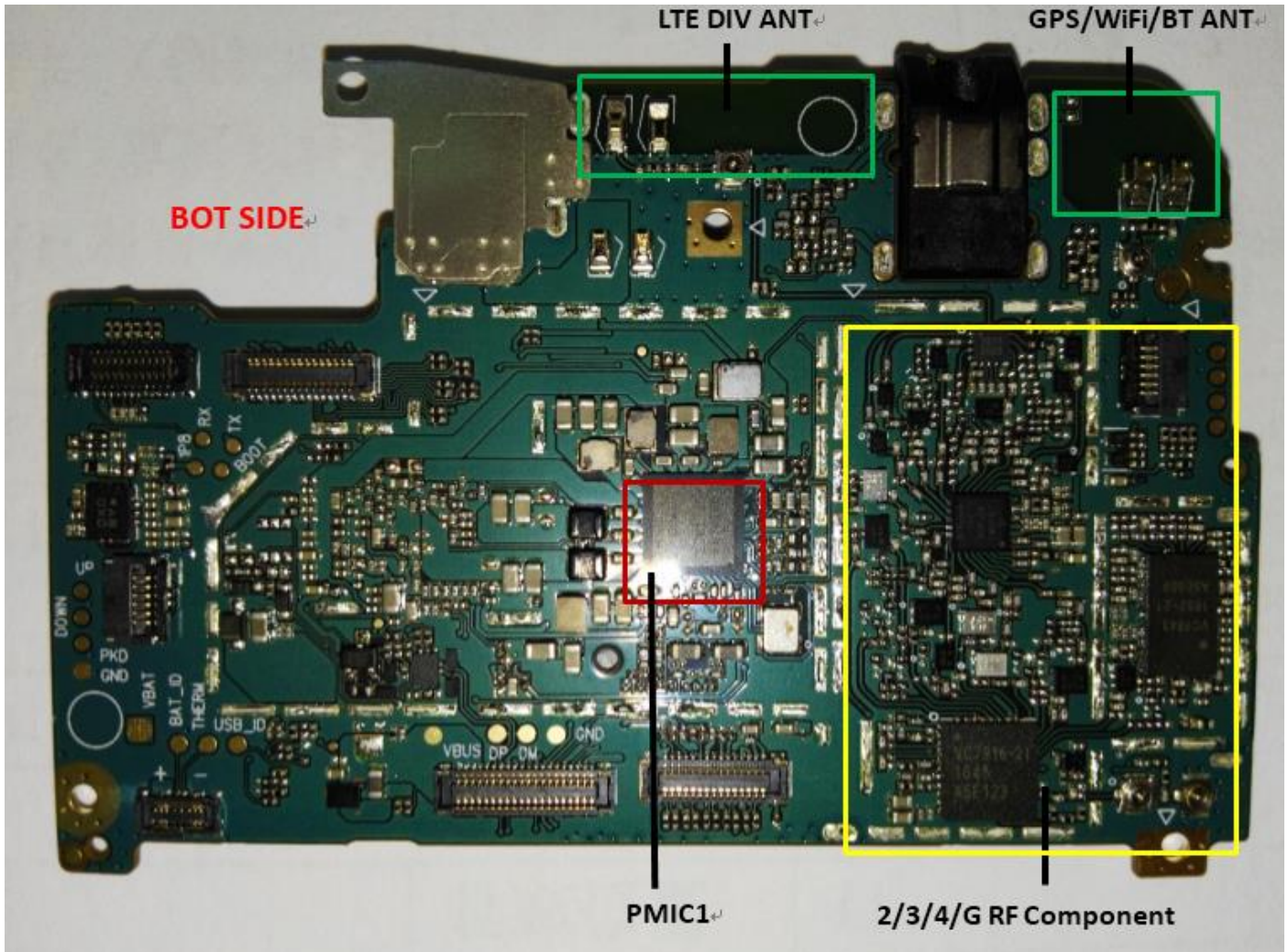
# ASUS ZC554KL A

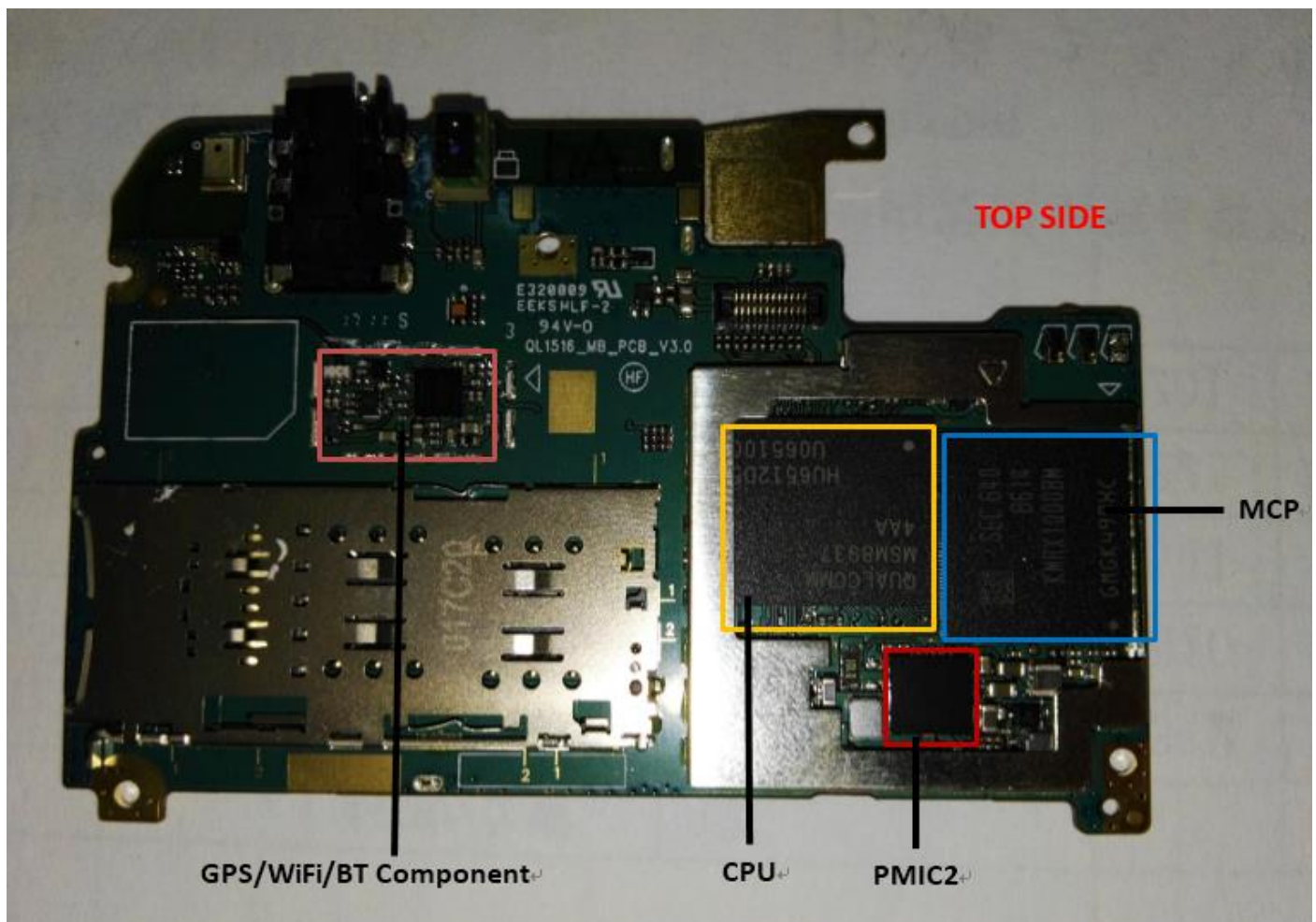
## RF Hardware Troubleshooting Guide

# ASUS ZC554KL trouble shooting guide V1 -RF

## 1.RF Areas

The need to calibrate the range is as follows: “phone2G/3G/4G”、“MCP”





**\*Supported 2G/3G/4GBands for ZC554KL:**

## 2.GSM(2G)/WCDMA(3G)/LTE(4G)/CDMA RF

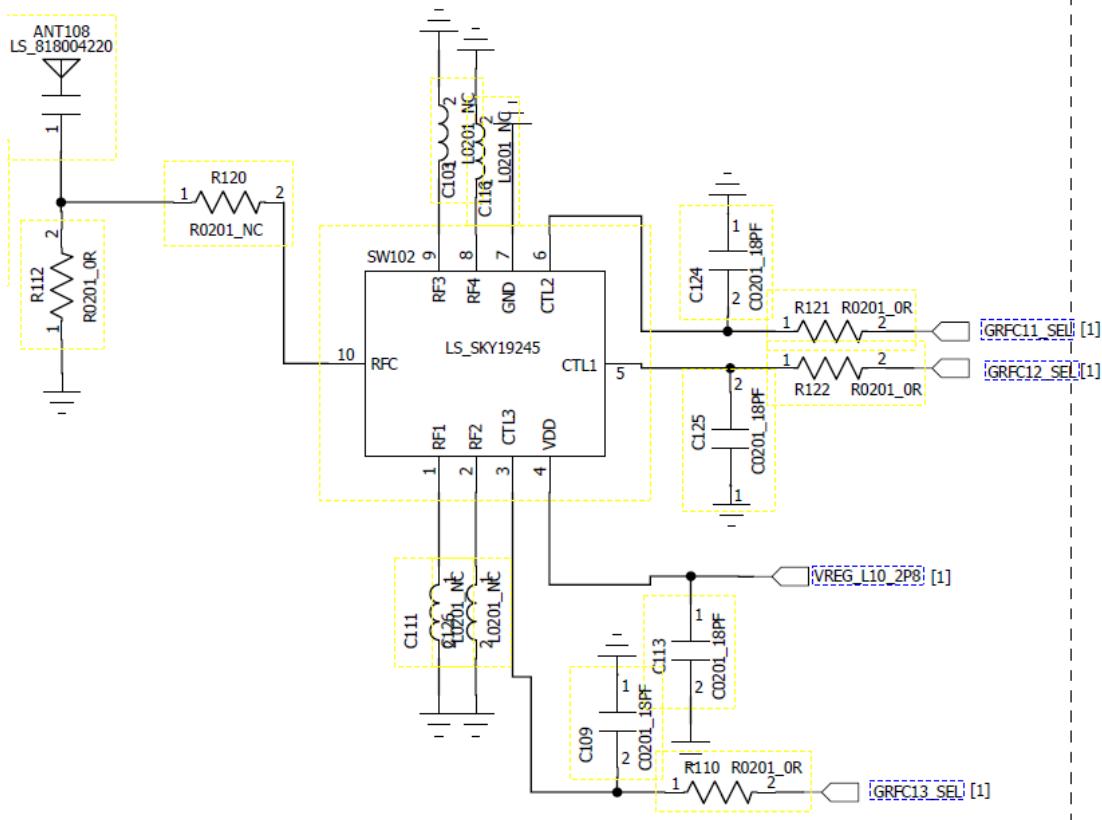
### 2.1 Circuit diagram

#### WW (WW+IN/ID)

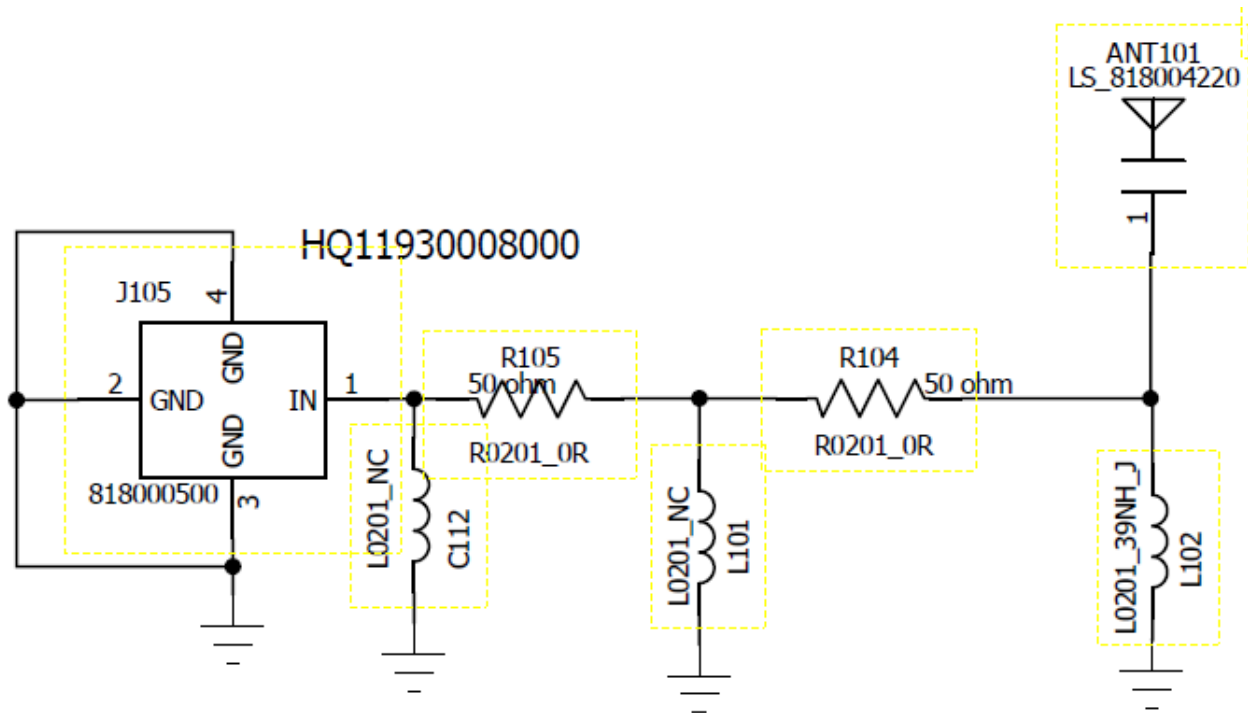
|                                                                              |                    |
|------------------------------------------------------------------------------|--------------------|
| 2G (EDGE/GPRS/GSM) :850MHz/900MHz/1800MHz/1900MHz                            | GSM(5/8/3/2)       |
| 3G (WCDMA ) :2100MHz(1)/850MHz(5)/900MHz(8)                                  | WCDMA(1/5/8)       |
| 4G :FDD-LTE: 2100MHz(1)/1800MHz(3)/850MHz(5)/2600MHz(7)/900MHz(8)/800MHz(20) | LTE (1/3/5/7/8/20) |
| 4G (TDD-LTE): 2300MHz(40)                                                    | B40                |

**Main ANT Tuner:**

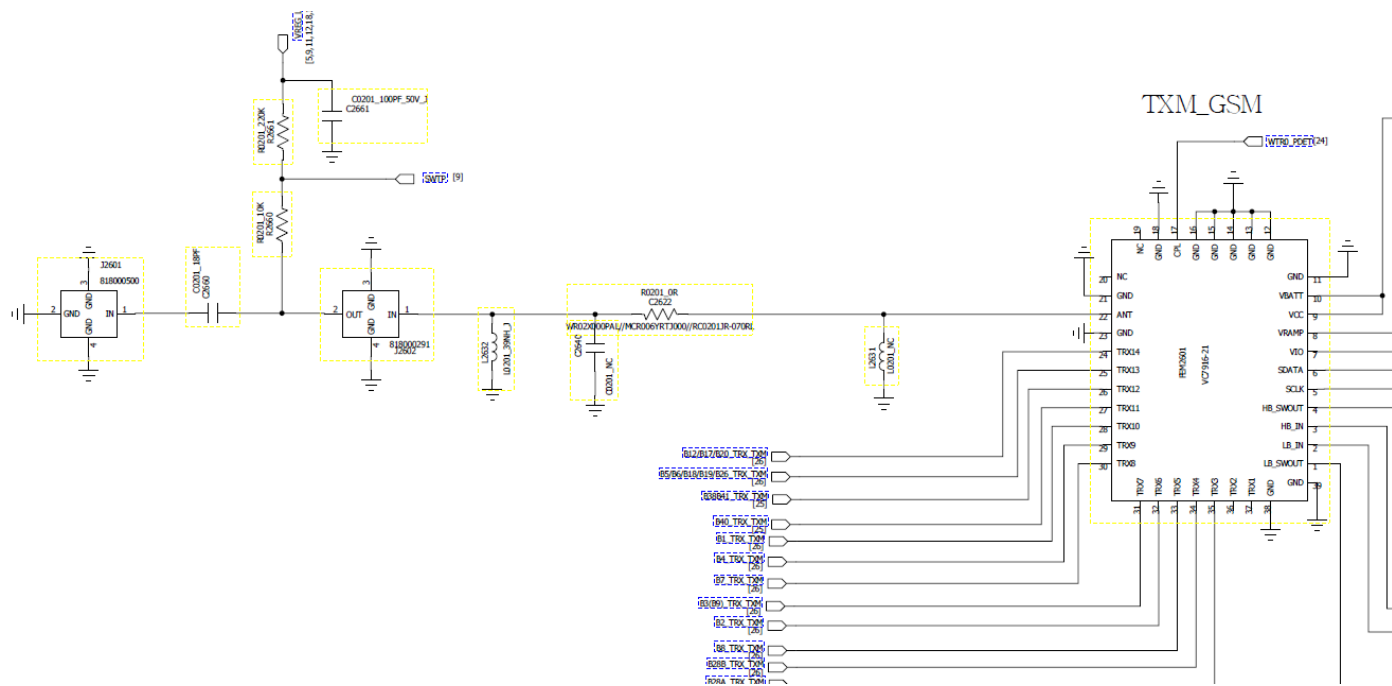




**Main ANT:**

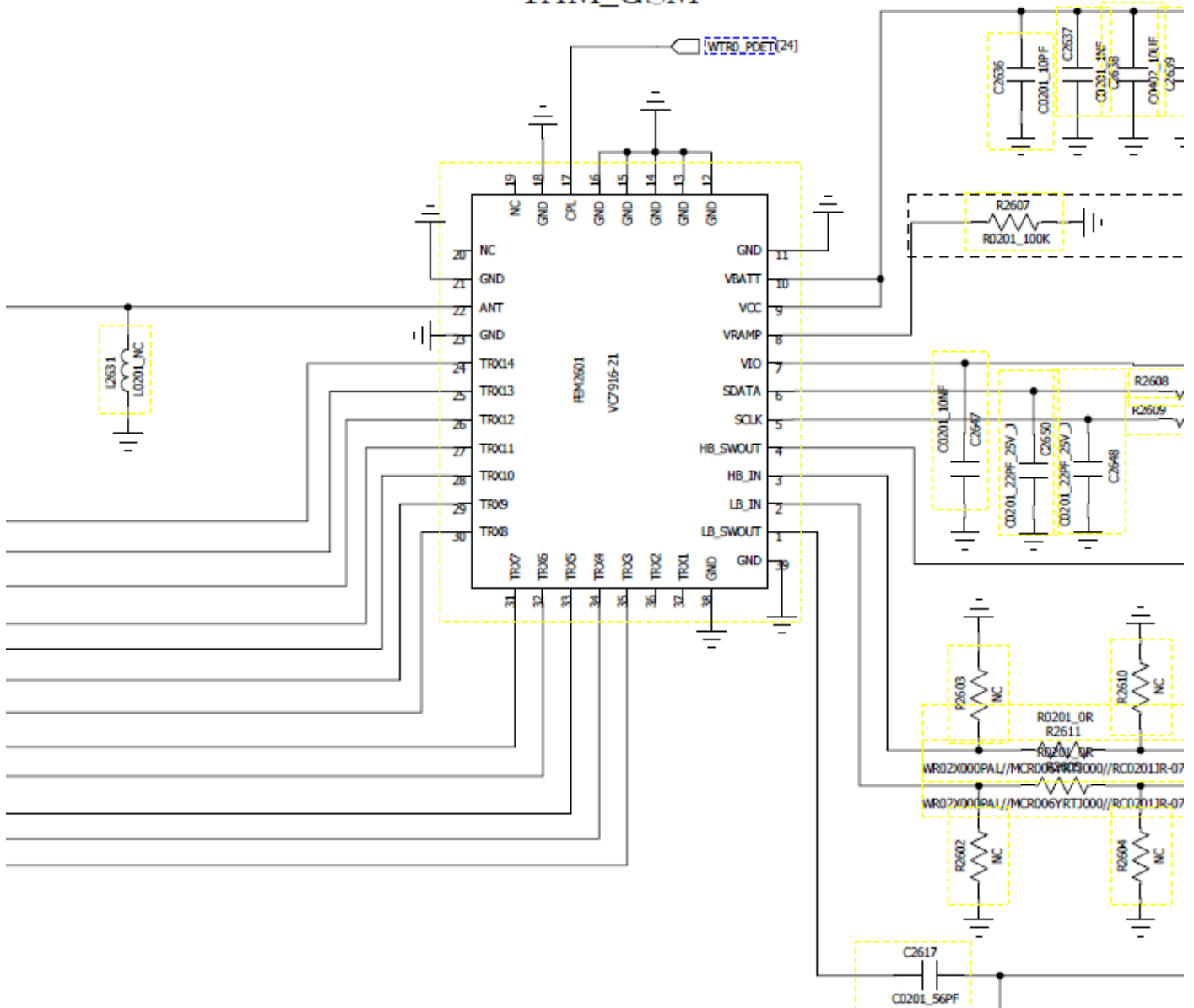


**RF Main SET 2/3/4G:**

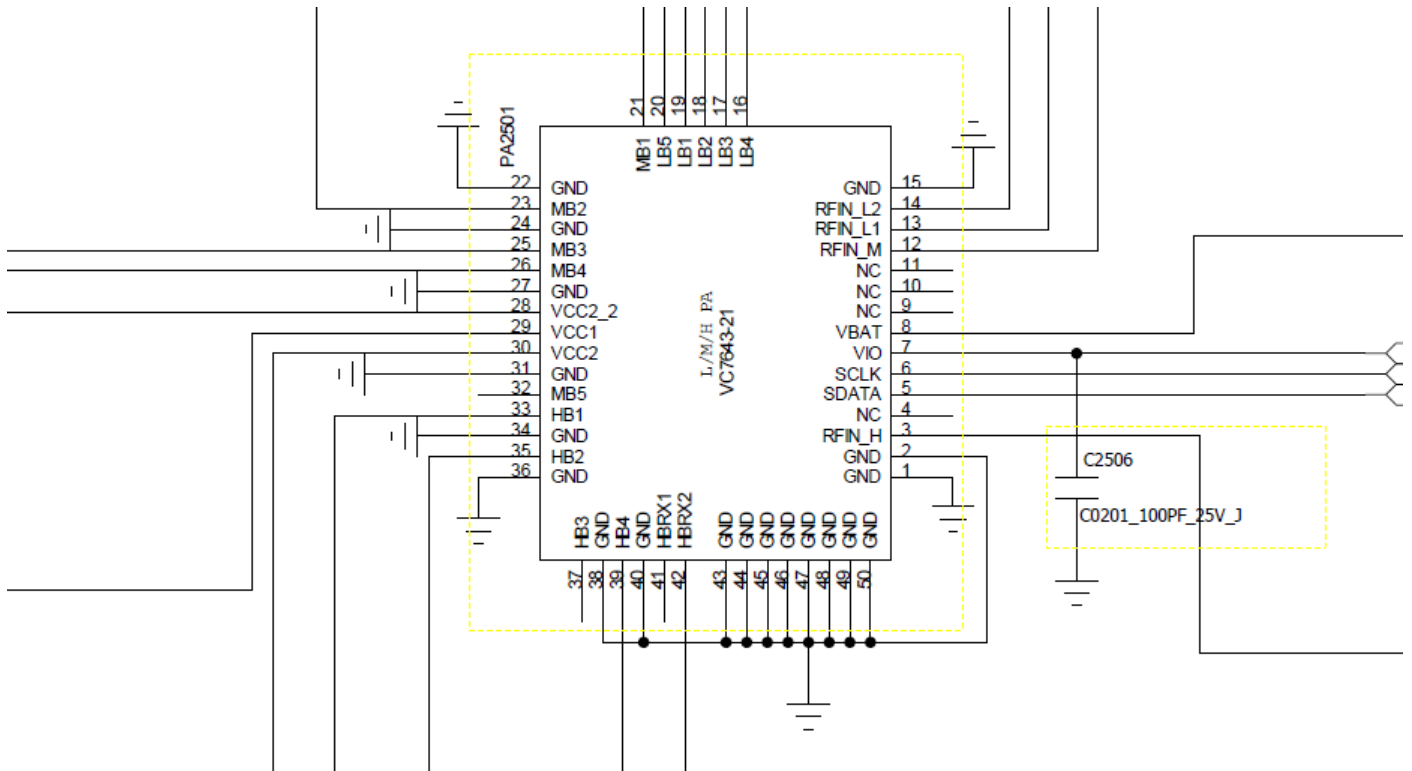


## GSM/TD-SCDMA PA+SP16T RF Switch:

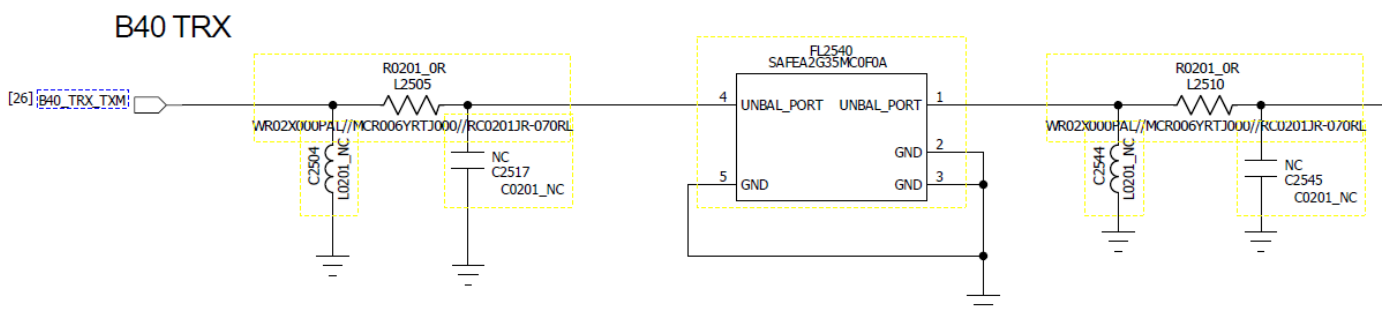
# TXM\_GSM



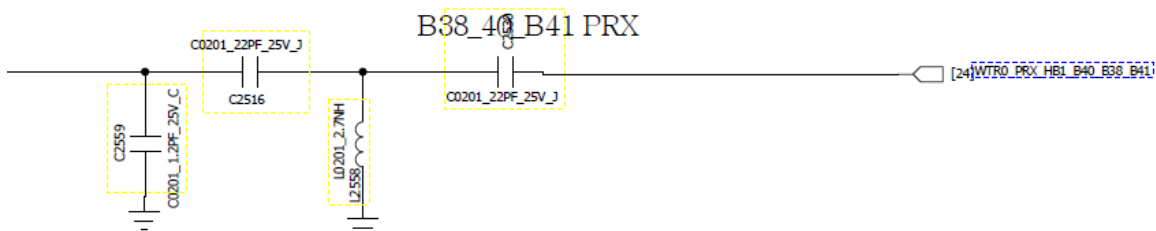
WCDMA+LTE PA:



## B40 TRX SAW:

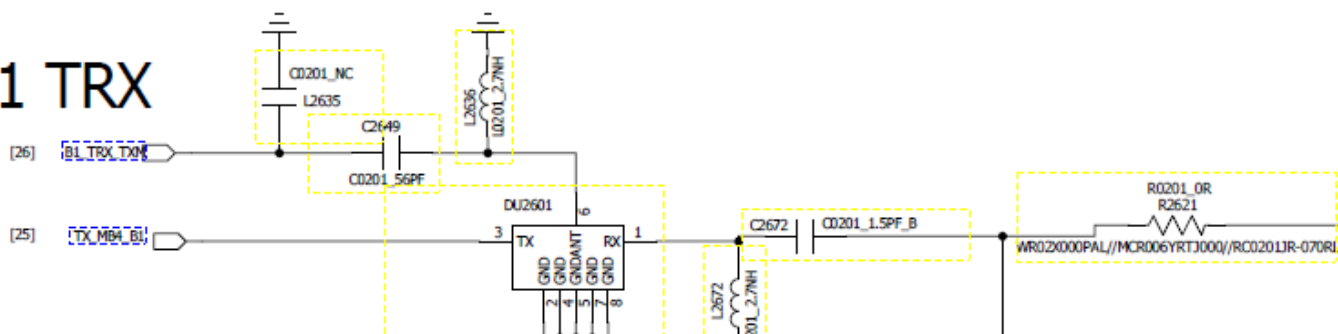


## B40 PRX:



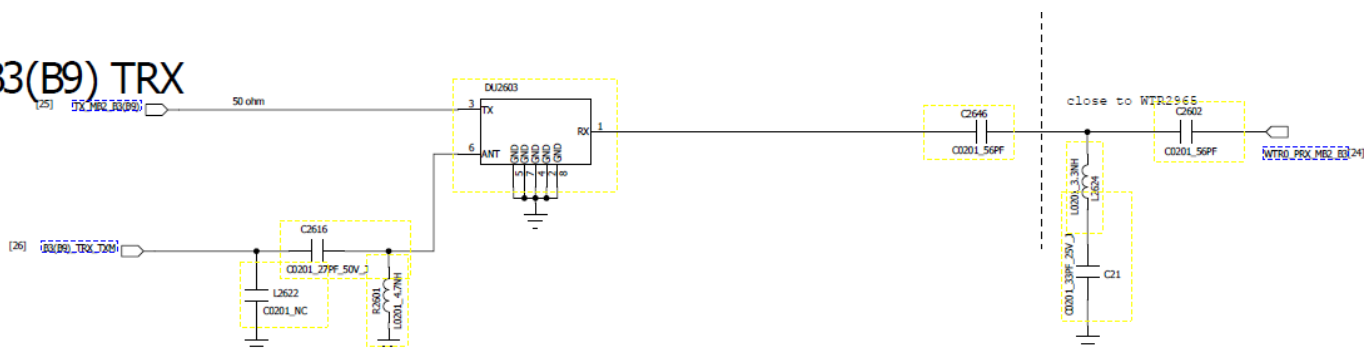
## B1 Duplexer:

## B1 TRX



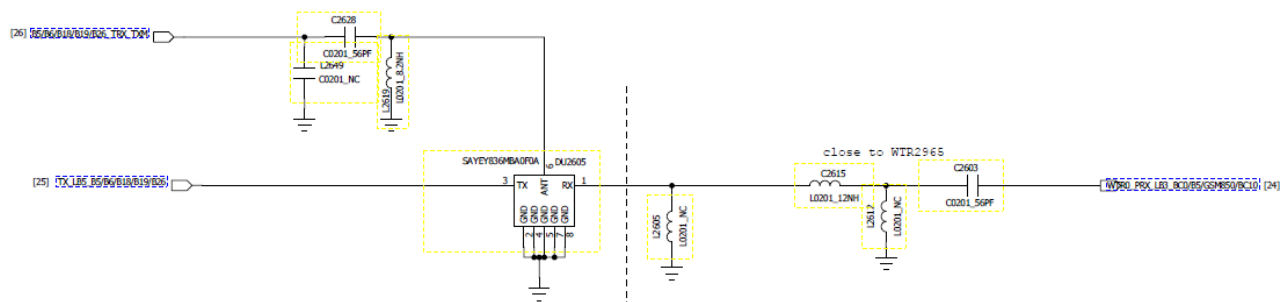
## B3 Duplexer:

### B3(B9) TRX



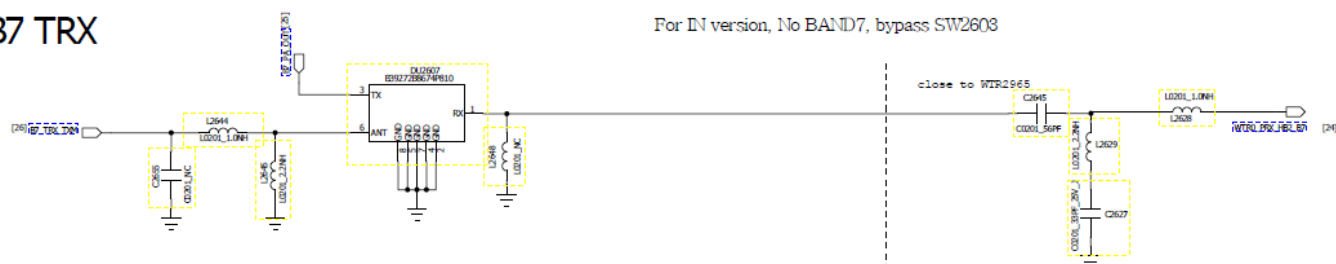
## B5/6/18/19/26 Duplexer:

### B5/B6/B18/B19\_TRX



## B7 Duplexer:

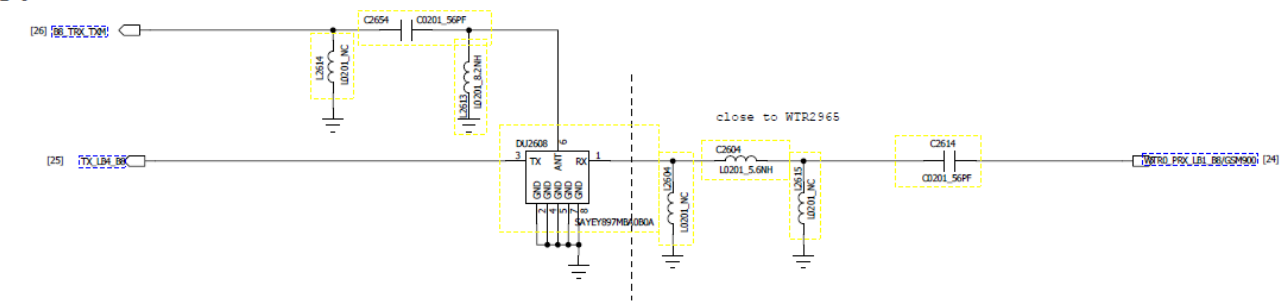
### B7 TRX



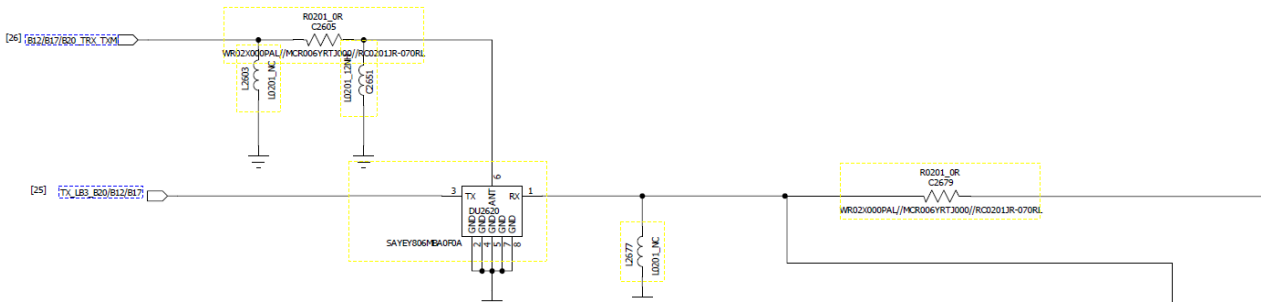
## B8 Duplexer:



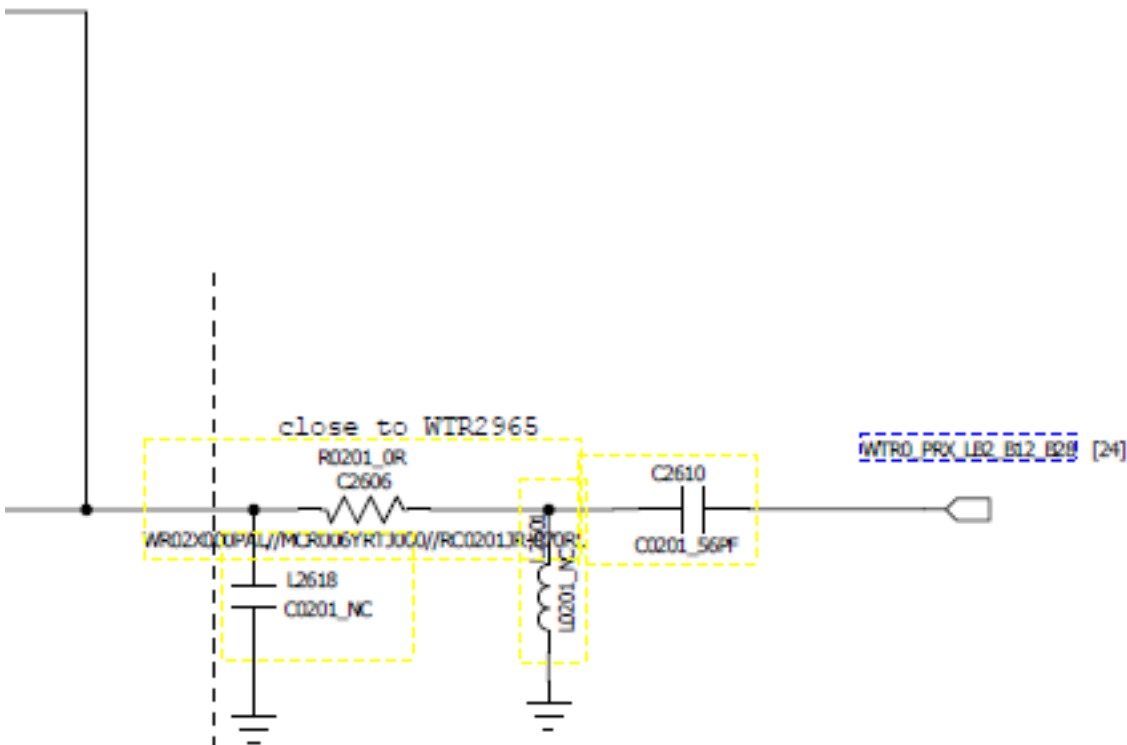
B8 TRX



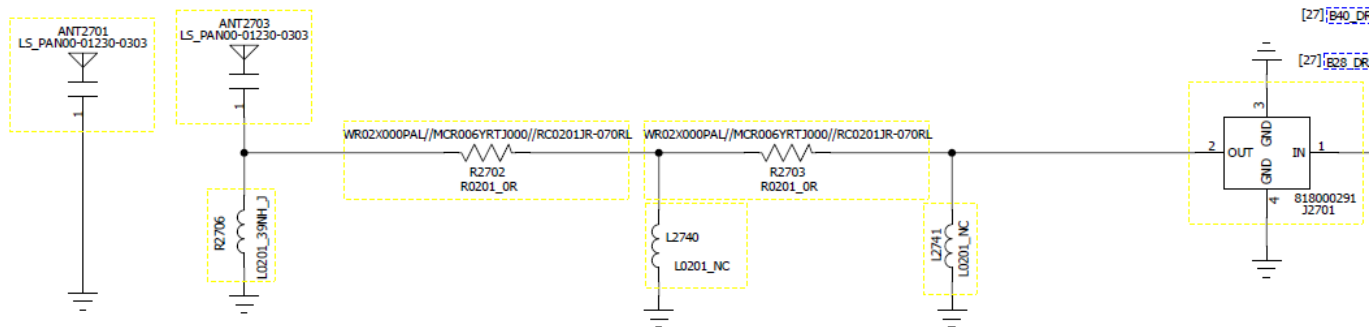
B20 Duplexer:  
B12/B17/B20 TRX



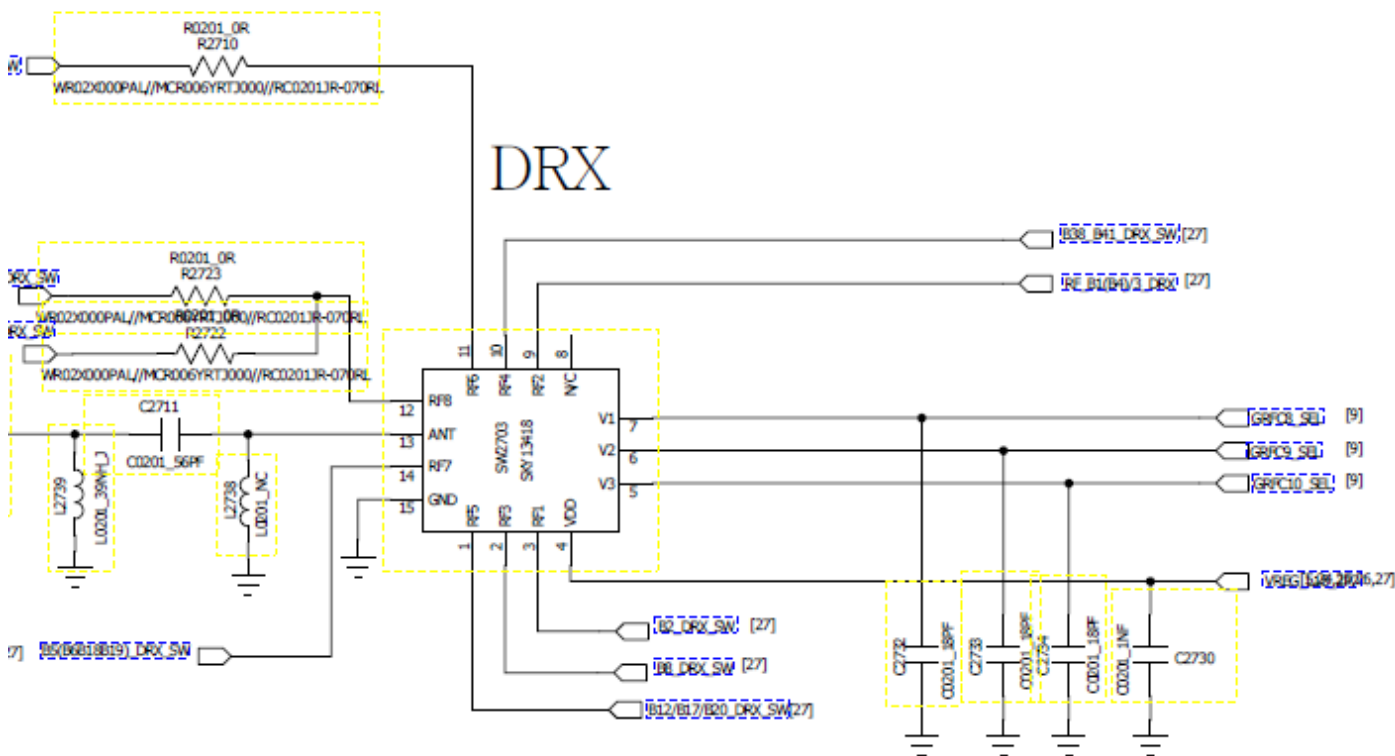
B20 PRX without switch:



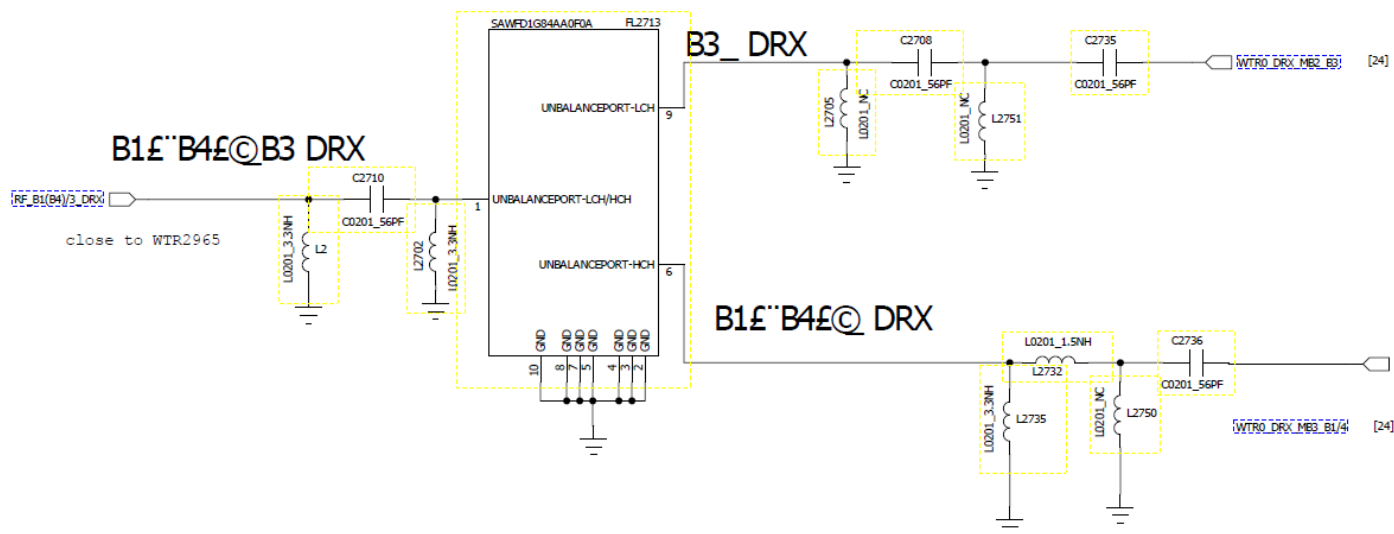
DRX RF SET:



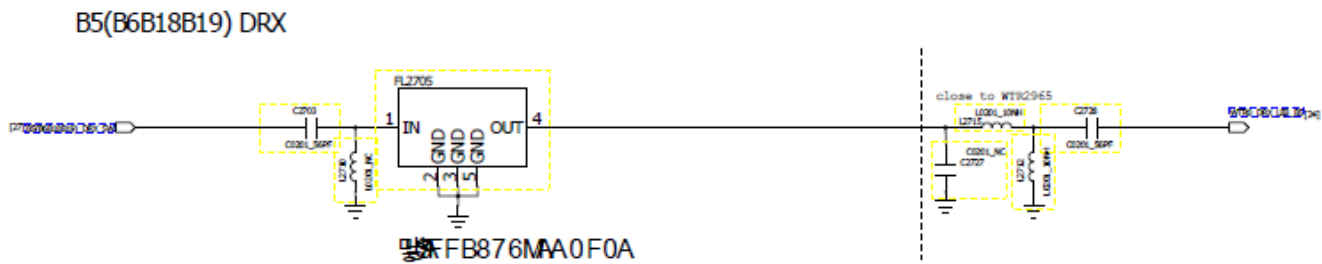
## DFX RF Switch:



## B1+B3 DRX SAW:



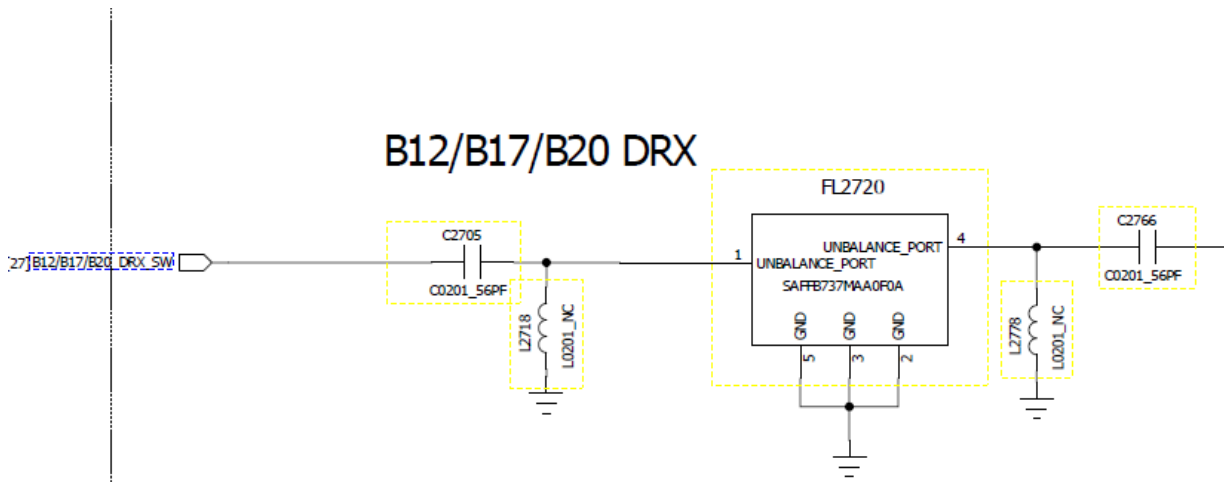
**B5/6/18/19/26 DRX SAW:**



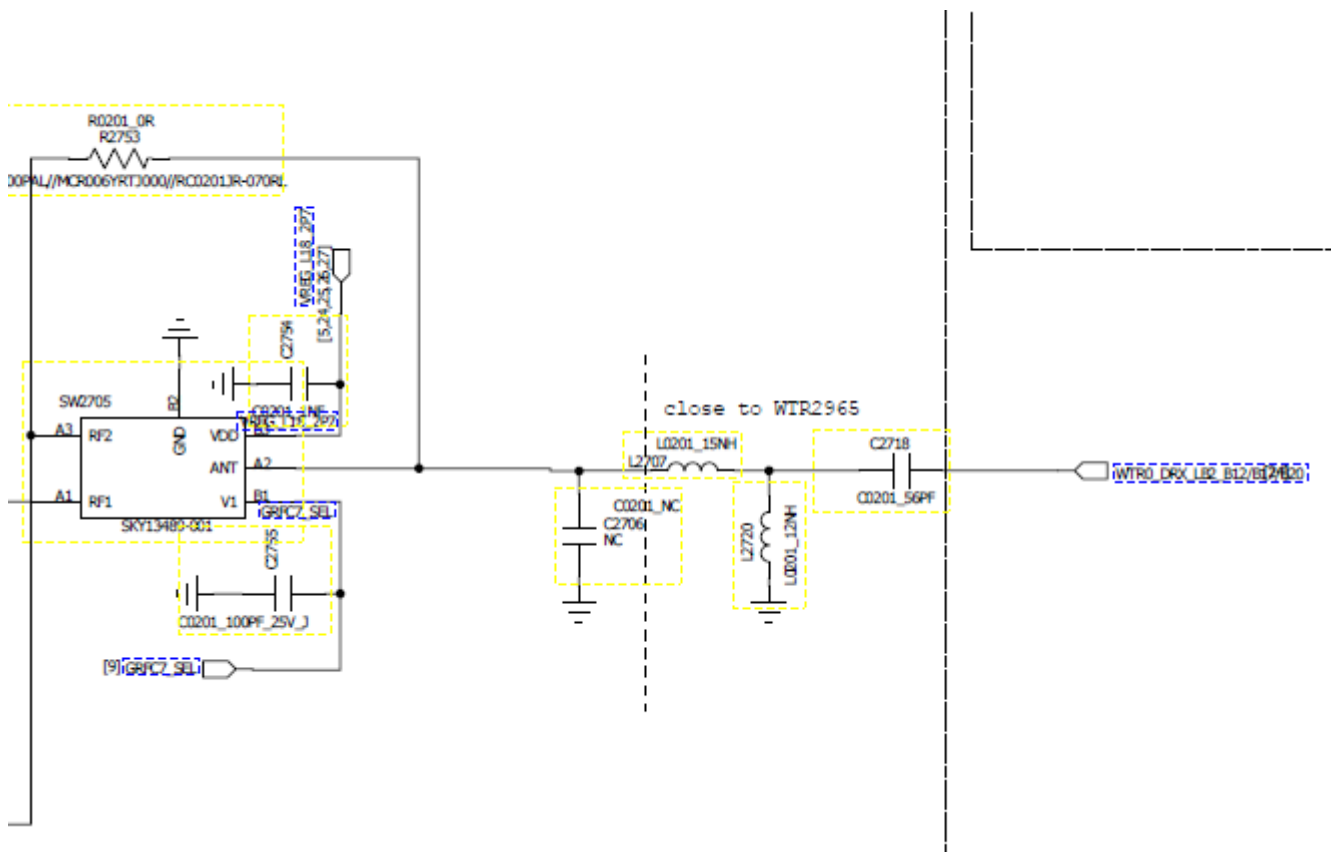
**B8 DRX SAW:**



**B20 DRX SAW:**

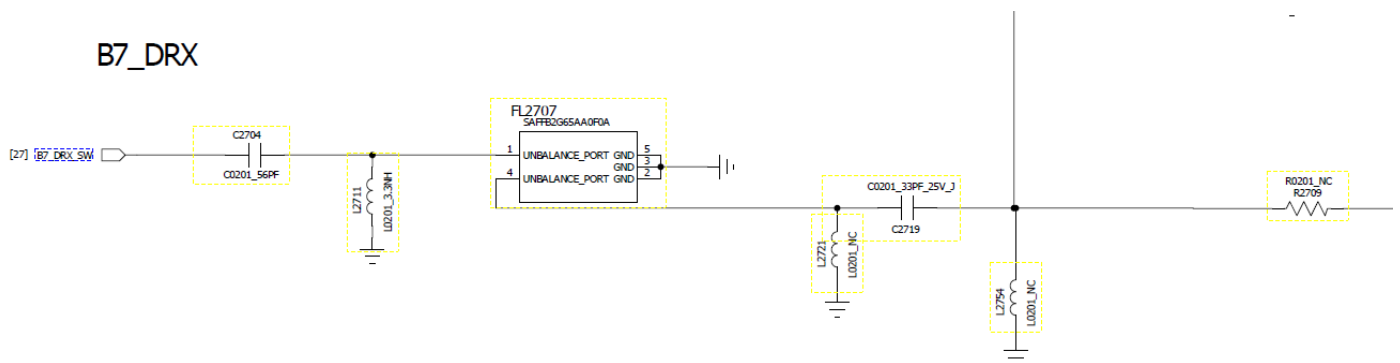


## B20 DRX without switch



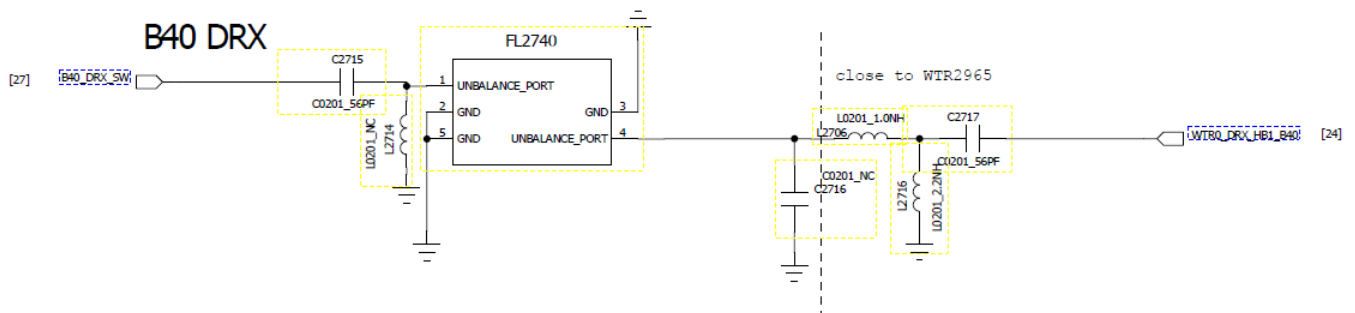
## B7 DRX SAW:

B7\_DRX

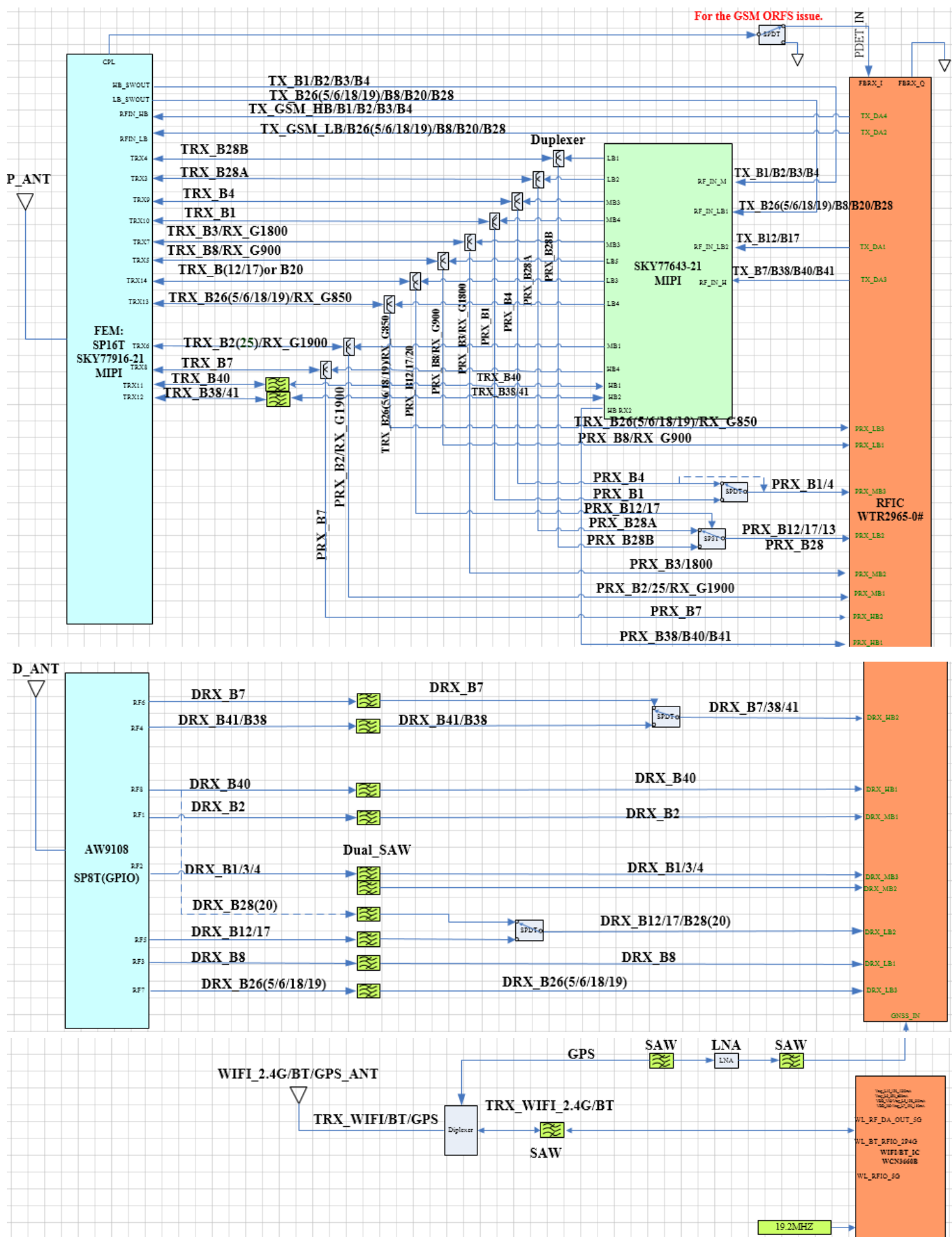


## B40 DRX:

B40 DRX



## 2.2RF Block Diagram:2G/3G/4G(GSM&CDMA&WCDMA&LTE)System



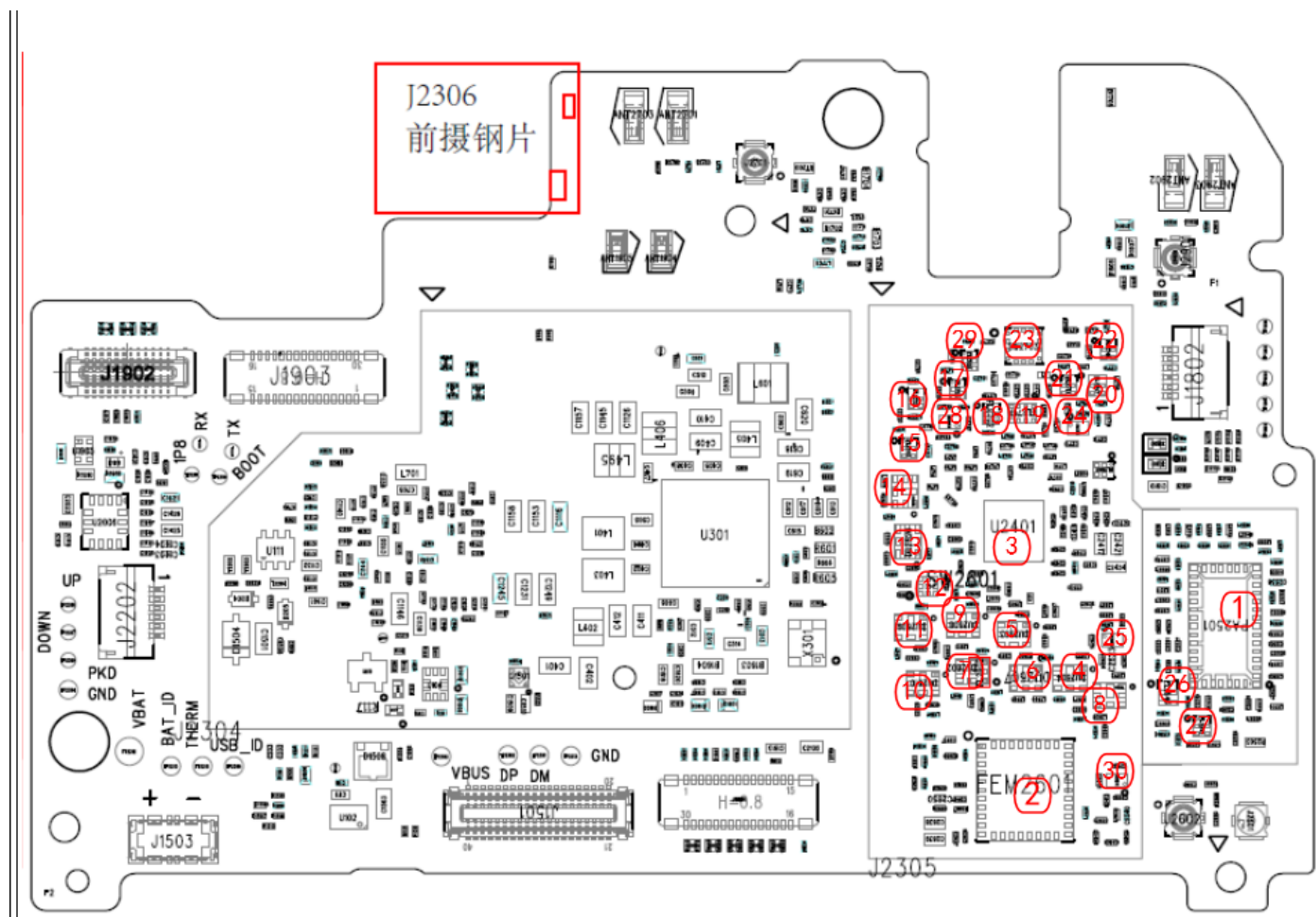


## 2.3 Key Components

### Electronic components description and circuit location code:

| 编号 | 位号      | 物料描述                                      |
|----|---------|-------------------------------------------|
| 1  | PA2501  | 射频放大器模组_LTE/TDD/FDD/MMPA                  |
| 2  | FEM2601 | 射频放大器_GSM/TD                              |
| 3  | U2401   | 射频收发器_CDMA/GSM/WCDMA/TD-SCDMA/TDD/FDD-TLE |
| 4  | DU2604  | NC                                        |
| 5  | DU2603  | 电磁干扰双向阻带滤波器_B3_单端                         |
| 6  | DU2607  | 电磁干扰双向阻带滤波器_B7_单端                         |
| 7  | FL2602  | 声表面滤波器_1960_RX                            |
| 8  | DU2601  | 电磁干扰双向阻带滤波器_band1_单端 1814                 |
| 9  | DU2608  | 电磁干扰双向阻带滤波器_band5_单端 1814                 |
| 10 | DU2610  | NC                                        |
| 11 | DU2609  | NC                                        |
| 12 | SW2601  | NC                                        |
| 13 | DU2620  | 电磁干扰双向阻带滤波器_B20/Unbalanced_1814           |
| 14 | DU2605  | 双工器_B8_单端_1814                            |
| 15 | FL2705  | 声表面滤波器_B8_RX                              |
| 16 | FL2720  | 声表面滤波器_B20/单端_1109                        |
| 17 | FL2704  | NC                                        |
| 18 | FL2702  | NC                                        |
| 19 | FL2713  | 声表面滤波器_B1/B3_RX                           |
| 20 | FL2707  | 声表面滤波器_B7/单端_1109                         |
| 21 | FL2741  | NC                                        |
| 22 | FL2740  | 声表面滤波器_B40/Rx/ub_1109                     |
| 23 | SW2703  | 射频开关_SP8T                                 |
| 24 | SW2701  | NC                                        |
| 25 | SW2605  | NC                                        |
| 26 | FL2540  | TDD-LTE_B40_TRX SAW                       |
| 27 | FL2542  | NC                                        |
| 28 | SW2705  | 射频开关_SPDT                                 |
| 29 | FL2708  | 声表面滤波器_GSM850_RX1109                      |
| 30 | SW2401  | 射频开关_SPDT                                 |

**Schematic diagram on the motherboard:**



### 3.WIFI/GPS/BT

### 3.1Circuit diagram

